Week 1: 17th may to 23rd may

Agenda

Genomic DNA extraction of S. cerevisiae (write strain)

Instruments required

- Autoclave
- Magnetic stirrer with heat plate
- Centrifuge
- PCR tubes and microcentrifuge tubes (1.5 ml, 0.5 ml)
- Gel electrophoresis apparatus

Reagents

- MQ / AMQ (ultrapure water)
- YPD Broth (Dextrose, Protease peptone, Yeast extract)
- YPD Agar
- Triton-SDS
- Phenol-chloroform isoamyl (25:24:1)
- TE buffer (pH 8.0)
- Glass beads
- Absolute ethanol and 70% ethanol
- 4M Ammonium Acetate
- 10X TBE buffer and Agarose powder
- AZ dye

18th may:

The following materials were autoclaved: pipette tips (1000 µl, 200 µl, and 10 µl), micro-centrifuge tubes (2 ml, 1.5 ml), 5 ml LB tubes (10), 100 ml LB flasks (4), and 300 ml of LA medium, YBD medium.

19th may:

Streaking of BY4741: BY4741 was streaked onto a plain YPD agar plate using a sterile spreader from the glycerol stock. The plate was kept in the incubator for growth.

20th may:

Inoculated S. cerevisiae twice from different colonies in 5 ml YPD and incubated at 37 ℃, 200 rpm for O/N (overnight)

21st may:

Observation

The growth of S. cerevisiae was optimal, with the culture being turbid.

Genomic DNA extraction

I. Growth of Cells

- Centrifuge at 13000 rpm for 15 seconds, then add in MCT.
- Remove the SN (supernatant) and add 1ml sterile H2O (AMQ outside purpose labelled), and resuspend cells by gentle vortex.
- Centrifuge at 13000 rpm for 15 seconds.
- Remove the SN and resuspend in 1ml of sterile H2O.
- Centrifuge at 13000 rpm for 15 seconds at RT (Room Temperature) and remove SN.

II. Extraction of DNA

- Add 0.2ml of Triton/SDS to cells (located in stand PB down table); do not double press the pipette.
- Add 0.2 ml of phenol:chloroform (add from the bottom layer) and 0.2gm of glass beads, then vortex for 30 seconds at RT.
- Add 0.2ml of TE (pH 8.0) (located in stand to the right PB down table) and mix by vortexing for 30 seconds.
- Separate the organic and aqueous phases by centrifuging at 13000 rpm for 10 minutes at RT.
- Transfer the aqueous layer to a new MCT without disturbing materials at the interface.

III. Isolation of DNA

- Add 700 µl of chilled (-20°C temp store) 100% ethanol and 10 µl (4M) Ammonium acetate to the aqueous solution and mix by inversion (5-10 times).
- Incubate at -20°C for 12 minutes (Enzyme fridge).
- Collect the precipitate of DNA by centrifuging at 13000 rpm for 10 minutes at RT, then discard the SN.
- Observe the pellet and add 0.5ml of chilled 70% ethanol.
- Centrifuge again at 13,000 rpm for 10 minutes.
- Discard SN, tap on tissue paper, and let it pellet dry at 60°C for 3 minutes with the lid of the MCT open (ensure no ethanol smell).
- Dissolve the pellet in 30 µl of AMQ at 60°C for 15 minutes (close the lid of the MCT and keep).

Important Additional Note

- After the air drying the pellet in the 70% ethanol step, always confirm drying by smelling.

- There should not be ethanol; a drop of water is fine, but not ethanol.

Gel electrophoresis preparation

- Prepared a 1:100 working stock of AZ dye.
- Prepared 1X TBE working solution (30 ml 10X TBE + 270 ml MQ).
- In a 250 ml flask, added 32 ml 1X TBE + 0.3 g agarose powder (10% gel). Heated for 1 minute, swirled until the temperature was bearable, added 10 µl of AZ dye working stock, and poured into the plate.

Lane Designation	**Content Loaded**	**Amount Loaded**
Vial M / D	Sample + AZ Dye + MQ	2 µl + 2 µl + 2 µl
Ladder	Ladder + AZ Dye + MQ	2 µl + 2 µl + 2 µl
Control	PCR Product + AZ Dye	5 µl + 2 µl

Observation

The gDNA extraction was successful. Sample M showed a clear, intact band, while sample D showed signs of degradation.

22nd may:

No work carried out — on hold due to unavailability of primers (received on 25th May).

23rd may:

No work carried out — on hold due to unavailability of primers.

24th may:

No work carried out — on hold due to unavailability of primers.

Conclusion

Media preparation, streaking, and inoculation of S. cerevisiae (BY4741) were completed successfully, and the culture reached optimal, turbid growth. Genomic DNA extraction was carried out and confirmed by gel electrophoresis; sample M yielded a clean, intact band, while sample D showed signs of degradation. Further experiments were paused from 22nd to 24th May due to the unavailability of primers required for the next stage of the project.

Week 2: 25.05.2025 to 30.05.2025

Agenda

Extraction of gDNA from BY4741, PCR amplification of the ORF and 3' flank regions of the gene of interest using a temperature gradient, troubleshooting of extraction/PCR failures, and construction of an intermediate plasmid (3' flank ligated into p426GPD).

Instruments required

- PCR machine
- Water bath
- Centrifuge
- Microwave
- Magnetic stirrer
- Electrophoresis tank with power supply
- UV Transilluminator
- Spectrophotometer / Nanodrop

Reagents

[Full reagent/recipe list to be added — TBE prep, AZ dye dilution, PCR reagents, digestion/ligation buffers, antibiotic stocks]

(25.05.2025)

PCR mix preparation

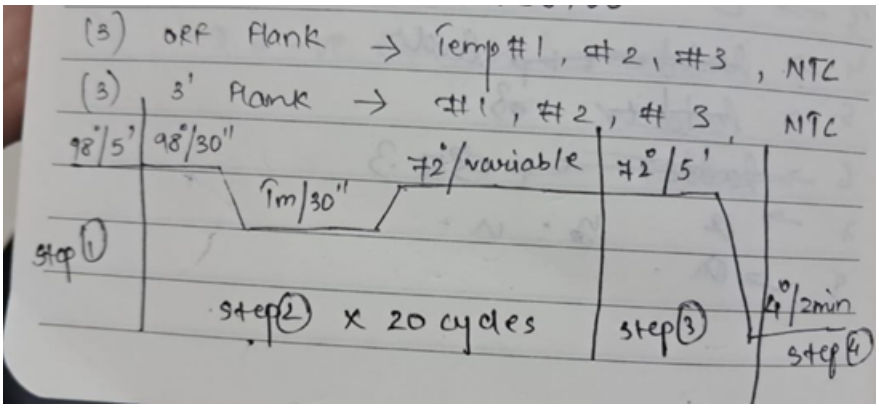
The PCR mix was prepared in the following amounts:

PCR for DNA Mutation and Amplification

Component	Volume
AMQ (ultrapure water)	10 µl
10X Buffer	2 µl
1mM dNTPs	2 µl
5µM Forward Primer (FP)	2 µl

5µM Reverse Primer (RP)	2 µl
gDNA (1:20 working stock)	2 µl
Enzyme	0.2 µl
Total Volume	**20 µl**

Gradient PCR set up

- ORF: Tm 56, 58, and 60 °C; extension (step 4) 2.5 min
- 3' flank: Tm 60, 62, and 64 °C; extension (step 4) 40 sec
- 

Gel electrophoresis of the PCR products was carried out to visualise amplification.

Observation

The gel could not be properly assessed, and results from this run were inconclusive.

(26.05.2025)

gDNA extraction — Attempt 1 and Attempt 2

The gDNA extraction protocol (as performed on 19.05.2025) was repeated independently, twice, in parallel:

1. Centrifuge at 13,000 rpm for 15 sec (2 MCTs of 0.5 ml each).
2. Remove the media and replace with 1 ml sterile water; resuspend the pellet by gentle vortexing.
3. Centrifuge at 13,000 rpm for 15 sec; discard supernatant.
4. Add 0.2 ml Triton-SDS, 0.2 ml phenol-chloroform-isoamyl alcohol (25:24:1), 0.2 ml TE (pH 8.0), and 0.2 g glass beads to the cells; resuspend by gentle pipetting.
5. Vortex at 1,400 rpm for 2 min.
6. Centrifuge at 13,000 rpm for 10 min for phase separation.

7. Transfer the aqueous layer to a new MCT.
8. Add 0.8 ml absolute ethanol + 10 µl 4M Ammonium Acetate; mix by inversion; incubate at -20°C for 10 min.
9. Centrifuge at 13,000 rpm for 10 min; discard supernatant.
10. Rinse the pellet with 0.5 ml of 70% ethanol.
11. Centrifuge at 13,000 rpm for 10 min; discard ethanol.
12. Air dry the DNA pellet for 5 min; add 50 µl AMQ.

Observation

Neither attempt yielded a visible DNA pellet of usable quality. Both extractions were deemed unsuccessful, and it was concluded that the gDNA was not viable for downstream PCR use.

PCR (repeated)

Since the gel from 25.05.2025 could not be assessed, the PCR reaction (same mix composition and gradient conditions as above) was repeated using the existing gDNA stock.

Observation

This PCR also failed to give results — no visible amplification bands were observed on the gel.

(27.05.2025)

Gradient PCR (repeated, with two gDNA sources)

The gradient PCR was repeated using the same PCR mix composition and cycling conditions as on 25.05.2025 (ORF: Tm 56/58/60°C; 3' flank: Tm 60/62/64°C), using two separate gDNA templates:

- gDNA extracted on 19.05.2025
- gDNA obtained from [Lab Name]

Observation

This run gave usable results, allowing the optimal annealing temperature to be narrowed down. However, with the 19th May gDNA, ORF amplification did not show distinct, well-amplified bands, indicating this gDNA sample remained of poor quality.

(28.05.2025)

PCR (repeated, larger pool)

A larger-volume PCR pool was set up and the reaction repeated for the 3' flank fragment at its optimized annealing temperature:

[Optimized annealing temperature (Tm) to be added]

Primary inoculation of DH5α (for plasmid isolation)

In the evening, DH5α was inoculated for subsequent plasmid isolation:

- A single colony of DH5α was inoculated into 5 ml LB broth.

- Incubated overnight at 37°C with shaking.

(29.05.2025)

Plasmid isolation of p426GPD from DH5α

Plasmid isolation was carried out using a plasmid isolation kit from GeneAll. The procedure followed was:

13. 1 ml of the bacterial culture was transferred into a 1.5 ml microcentrifuge tube and centrifuged at 13,000 g for 1 minute; supernatant discarded.
14. This step was repeated five times for the entire 5 ml culture, collecting all cell pellets in a single tube.
15. 250 µl of S1 buffer was added to the pellet and resuspended thoroughly using a pipette.
16. 250 µl of S2 buffer was added; the tube was inverted four times to mix gently and allowed to rest for 1 minute.
17. 350 µl of S3 buffer was added; the tube was inverted four times and left to stand for 30 seconds.
18. Centrifuged at 11,000 g for 10 minutes; allowed to rest for 10 minutes, then centrifuged again at 11,000 g for 5 minutes.
19. The supernatant was collected carefully onto a filter column and centrifuged at 11,000 g for 1 minute; flow-through discarded.
20. 500 µl of AW buffer was added and centrifuged at 11,000 g for 1 minute; flow-through discarded.
21. 700 µl of PW buffer was added and centrifuged at 11,000 g for 1 minute; flow-through discarded.
22. The column was transferred to a new 1.5 ml microcentrifuge tube.
23. 50 µl of EB buffer was added dropwise directly onto the membrane; left to stand for 1 minute, then centrifuged at 11,000 g for 1 minute.
24. The eluted plasmid DNA was stored at 4°C for further use.

Purification of vector (p426GPD) and 3' flank region

[Purification process to be added]

Restriction digestion of 3' Flank and p426GPD vector

Two separate reactions were set up:

Component	Tube 1: S2 (3' Flank)	Tube 2: p426GPD Vector
DNA Template	90 µl (PCR Product)	10 µl (Plasmid)
10X NEB Buffer	20 µl	5 µl
SacI Enzyme	2 µl	2 µl

Milli-Q Water	88 µl	33 µl
Total Volume	200 µl	50 µl

- Step 1 — SacI digestion: Vials briefly spun to collect liquid, then incubated at 37°C for 90 minutes, with a tap-and-spin every 30 minutes. Heat-inactivated at 65°C for 20 minutes.
- Step 2 — BamHI digestion: 2 µl BamHI added directly to each vial; incubated at 37°C for 60 minutes, with a tap-and-spin after the first 30 minutes.

Post-digestion column purification

Since BamHI is not easily heat-inactivated at 65°C (requiring 80°C, which risks DNA damage), the spin-column kit was used immediately following digestion:

- 5 volumes of NR Buffer added to each tube (1 ml for Tube 1; 250 µl for Tube 2); left to sit for 5 minutes.
- Applied to column and centrifuged at 12,100 rpm for 1 minute; flow-through discarded. Washed twice with 700 µl NW Buffer.
- Empty column centrifuged for 2 minutes at max speed to remove residual ethanol.
- Eluted with 30 µl EB Buffer, left to sit for 5 minutes, then centrifuged at 12,500 rpm to collect digested DNA.

Ligation

[Ligation reaction details to be added]

Bacterial transformation

25. Ligation mixture combined with 100 µl competent cells (thawed on ice); mixed by gentle swirling.
26. Incubated on ice for 30 minutes.
27. Heat shock given on a heat block for exactly 90 seconds.
28. Placed on ice for 5 minutes.
29. 1 ml sterile LB added; incubated in a shaker incubator for 45 minutes.
30. Centrifuged for 2 minutes at 13,000 rpm; supernatant discarded.
31. 100 µl AMQ added and vortexed until resuspended.
32. Plated onto LA + Ampicillin plate.
33. Incubated overnight at 37°C.

In the evening, the transformation plate was streaked.

(30.05.2025)

Holiday — no lab activity.

Conclusion

The gDNA extracted on 19.05.2025 was likely degraded or of insufficient purity, as repeated extraction attempts failed and PCR amplification using this template was inconsistent. Externally sourced gDNA [from Lab Name] gave more reliable amplification and was used to narrow down the optimal annealing temperature going forward.

Week 3: 1st June to 5th June

Agenda

- PCR amplification of the 3′ flank and ORF of *S. cerevisiae*
- Gradient PCR to optimize amplification temperatures
- Genomic DNA extraction of *S. cerevisiae*
- Plasmid isolation of p426GPD from *E. coli*
- Restriction digestion of p426GPD vector and 3′ flank insert
- Gel electrophoresis and purification of PCR products
- Ligation of the 3′ flank into the p426GPD vector
- Transformation of the ligation product into competent *E. coli*

Instruments required

- Autoclave
- PCR machine
- Gel electrophoresis apparatus
- Microcentrifuge
- Water bath
- Shaker incubator
- Magnetic stirrer with heat plate
- Micropipettes
- PCR tubes
- 1.5 ml microcentrifuge tubes

Reagents

- AMQ
- dNTPs
- Forward and reverse primers
- *S. cerevisiae* genomic DNA
- PCR enzyme
- Phusion DNA polymerase

- 10X PCR buffer
- Agarose
- AZ dye
- TBE buffer
- p426GPD plasmid
- Buffer S1, S2 and S3
- Buffer AW and PW
- Mini Q column
- SacI
- BamHI
- 10X restriction digestion buffer
- GB buffer
- DNA ligase
- Ligase buffer
- Competent *E. coli* cells
- LA + ampicillin plates

1st June:

PCR amplification of the 3′ flank

PCR was performed to amplify the 3′ flank region of *S. cerevisiae*.

The reaction mixture used was:

Reagent	Volume
AMQ	10 µl
1 mM dNTPs	2 µl
5 µM Forward primer	2 µl
5 µM Reverse primer	2 µl
S. cerevisiae gDNA (1:20 dilution)	2 µl
PCR enzyme	0.2 µl

The PCR setup was prepared according to the PCR setup shown in the experimental record.

Gradient PCR

Gradient PCR was performed to optimize the annealing temperature for amplification of the target regions.

For the **ORF (2.2 kb)**, the following temperatures were tested:

- 56°C
- 58°C
- 60°C

For the **3′ flank (528 bp)**, the following temperatures were tested:

- 60°C
- 62°C
- 64°C

Observation

The PCR was found to be unsuccessful after performing a gel electrophoresis run. No satisfactory amplification of the desired products was observed.

Primary inoculation of *S. cerevisiae*

A primary inoculation of *S. cerevisiae* was performed to obtain fresh culture for another attempt at genomic DNA extraction.

Primary inoculation of *E. coli* containing p426GPD

A primary inoculation of *E. coli* containing the p426GPD plasmid was also performed for subsequent plasmid isolation.

2nd June:

Plasmid Isolation of p426GPD

Plasmid isolation was performed from the *E. coli* culture containing p426GPD.

I. Cell Lysis

1. The bacterial culture was centrifuged at **13,000 rpm for 2 min** to pellet the cells.
2. The supernatant was discarded.
3. The pellet was thoroughly resuspended in **250 µl Buffer S1** containing RNase A.
4. **250 µl Buffer S2** was added and the tube was gently inverted 4–5 times without vortexing.

5. The sample was incubated until the lysate became clear, ensuring that the incubation did not exceed 5 min.
6. **350 µl Buffer S3** was added and the tube was immediately inverted 4–6 times without vortexing.
7. The lysate was centrifuged at **13,000 rpm for 10 min**.

II. DNA Purification

1. The cleared supernatant was carefully transferred to a mini Q column placed in a collection tube.
2. The column was allowed to stand for **5 min** to maximize DNA binding.
3. The column was centrifuged at **13,000 rpm for 30 s** and the flow-through was discarded.
4. **500 µl Buffer AW** was added, followed by centrifugation for 30 s. The flow-through was discarded.
5. **700 µl Buffer PW** was added, followed by centrifugation for 30 s. The flow-through was discarded.
6. The empty column was centrifuged at **13,000 rpm for 1 min** to remove residual wash buffer.
7. The spin column was transferred to a fresh 1.5 ml microcentrifuge tube.
8. **30 µl AMQ** was added directly to the centre of the membrane without touching the membrane bed.
9. The column was incubated for **5 min at room temperature**.
10. The column was centrifuged at **13,000 rpm for 1 min** to elute the plasmid DNA.

The eluate obtained contained the purified plasmid DNA.

Genomic DNA extraction

Genomic DNA extraction of *S. cerevisiae* was performed again.

PCR

Phusion PCR was performed for amplification of ORF and Taq polymerase was done for amplification of 3’ Flank.

Both PCR reactions were successful, and the amplified products were purified.

Gel electrophoresis

A gel electrophoresis run was performed using:

- ORF PCR product
- 3′ flank PCR product
- *S. cerevisiae* gDNA
- p426GPD plasmid

Observation

The gel confirmed successful amplification of both the ORF and 3′ flank PCR products.

The optimized temperatures were:

- **ORF: 58°C**
- **3′ flank: 60°C**

However, the genomic DNA extraction and p426GPD plasmid isolation were unsuccessful.

3rd June:

Media and buffer preparation

The following materials were autoclaved:

- TBE
- LB medium — 2 bottles

Genomic DNA extraction

Genomic DNA extraction of *S. cerevisiae* was performed again to obtain suitable template DNA for downstream experiments.

4th June:

Plasmid Isolation

Plasmid isolation of p426GPD was performed again.

Genomic DNA extraction

Genomic DNA extraction was performed again; however, the extraction was unsuccessful.

Restriction Digestion of Vector and Insert

Restriction digestion was performed on:

- **Vector:** p426GPD
- **Insert:** 3′ flank

Two 50 µl reactions were prepared for each component, giving a total reaction volume of 100 µl for the vector and insert.

Reagent	Vector	Insert
AMQ	76 µl	78 µl
10X buffer	10 µl	10 µl
SacI	2 µl	2 µl
BamHI	2 µl	2 µl
DNA	10 µl	8 µl
Total	**100 µl**	**100 µl**

The 100 µl reaction mixtures were divided into two tubes and incubated at **37°C for 30 min**.

Purification

The digested vector and insert were combined into a single 1.5 ml microcentrifuge tube.

300 µl GB buffer was added, followed by purification using the previously established purification procedure.

After purification, **2 µl of each product** was loaded onto a gel for electrophoresis.

Observation

The gel run showed that no visible vector or insert was obtained. This indicated that the plasmid isolation had failed and that sufficient vector and insert were not available for the next step.

A primary isolation of p426GPD was therefore initiated again.

5th June:

Plasmid Isolation

Plasmid isolation of p426GPD was performed again to obtain the vector for cloning.

Restriction Digestion

Restriction digestion of the following was performed:

- **Insert:** 3′ flank
- **Vector:** p426GPD

Gel electrophoresis

A gel electrophoresis run was performed after restriction digestion.

Observation

The gel run was successful, confirming the presence of both the digested **3′ flank insert** and **p426GPD vector**. The required insert and vector were therefore available for the ligation step.

Ligation

The digested 3′ flank insert was ligated into the p426GPD vector.

The ligation reaction consisted of:

Reagent	Volume
Vector	5 µl
Insert	6 µl
Ligase buffer	2 µl
Ligase enzyme	1 µl
AMQ	6 µl
Total	**20 µl**

The ligase buffer was completely thawed and vortexed to ensure that the salts were properly mixed.

After preparing the ligation mixture, the tube was gently tapped to mix and briefly centrifuged.

The reaction was incubated for **1.5 h**.

Transformation into competent *E. coli*

Transformation was performed using **100 µl competent cells** and the ligated 3′ flank-p426GPD construct.

1. The ligation product was added to the competent cells.
2. The mixture was gently swirled after addition.
3. The cells were incubated on ice for **30 min**.
4. Heat shock was performed at **42°C for 90 s** in a water bath.
5. The cells were immediately placed on ice for **5 min**.
6. **1 ml sterile medium** was added.
7. The cells were incubated in a shaker incubator at **37°C for 45 min**.
8. The transformation mixture was centrifuged for **2 min**.
9. The supernatant was discarded.
10. **100 µl AMQ** was added to the pellet and the mixture was vortexed.
11. **0.2 ml** of the transformed cell suspension was spread onto an LA plate containing ampicillin.
12. The plate was incubated at **37°C overnight**.

Conclusion

PCR amplification of the ORF and 3′ flank was successfully optimized, with **58°C identified as the optimum temperature for the ORF and 60°C for the 3′ flank**. Multiple attempts at *S. cerevisiae* genomic DNA extraction and p426GPD plasmid isolation were performed, with several unsuccessful attempts requiring repetition.

Restriction digestion initially failed due to unsuccessful plasmid isolation; however, a subsequent attempt successfully yielded both the p426GPD vector and 3′ flank insert. The purified vector and insert were then ligated, followed by transformation into competent *E. coli*. The transformed cells were plated on LA + ampicillin plates and incubated overnight for selection.Waited to see if transformation would be a success.

Week 4: 8th June to 12th June

Agenda

- Troubleshooting transformation and cloning
- Construction of the intermediate recombinant p426GPD plasmid containing the ORF
- Screening of recombinant colonies
- PCR amplification of the 3′ flank
- Addition of the 3′ flank to the intermediate recombinant plasmid

- Construction of the final recombinant plasmid
- Genomic DNA extraction
- Pure plasmid transformation

Instruments required

- PCR machine
- Gel electrophoresis apparatus
- Centrifuge
- Water bath
- Shaker incubator
- Autoclave
- Micropipettes
- PCR tubes and microcentrifuge tubes
- -20°C freezer

Reagents

- AMQ
- *S. cerevisiae* genomic DNA
- PCR reagents
- p426GPD plasmid
- Intermediate recombinant plasmid
- 3′ flank PCR product
- Restriction enzymes XhoI, KpnI, SacI and BamHI
- Restriction digestion buffer
- DNA ligase and ligase buffer
- Competent *E. coli*
- LB medium
- LA + ampicillin plates

8th June:

Transformation

The transformation performed previously failed. Since no 3′ flank insert was remaining, the cloning strategy was shifted to the **ORF**.

The restriction enzymes and other reagents were changed accordingly. Restriction digestion of the p426GPD vector and ORF insert was performed using **XhoI and KpnI**, followed by ligation and transformation into competent *E. coli*.

9th June:

Observation

Colonies were obtained after overnight incubation, indicating that the transformation was successful.

Two colonies were selected and used for primary inoculation.

PCR amplification of 3′ flank

PCR was performed again to amplify the **3′ flank** of *S. cerevisiae*. The PCR product was successfully obtained and purified for subsequent cloning into the intermediate recombinant plasmid.

10th June:

Plasmid Isolation

Plasmid isolation was performed from **six colonies**, designated:

- P1
- P2
- P3
- P4
- P5
- P6

These plasmids represented the **intermediate recombinant plasmid containing the ORF**.

Restriction Digestion Screening

P1 and P2 were screened by restriction digestion using XhoI and KpnI.

Reagent	Volume
AMQ	6 µl

10X buffer	1 µl
XhoI	0.5 µl
KpnI	0.5 µl
DNA	2 µl
Total	**10 µl**

Observation

The screening was successful. Two distinct bands corresponding to the **vector and ORF insert** were observed, confirming successful construction of the intermediate recombinant plasmid.

The remaining isolated plasmids were pelleted and stored at **-20°C** for future use.

11th June:

Addition of 3′ Flank

P1 was selected as the vector for addition of the 3′ flank.

Restriction digestion was performed on:

- **Insert:** 3′ flank
- **Vector:** Intermediate recombinant plasmid P1

The restriction enzymes **SacI and BamHI** were used with the newly prepared reagents.

The digestion was performed sequentially:

- SacI digestion — **45 min**
- BamHI digestion — **30 min**

Gel electrophoresis

The digested vector and 3′ flank were analyzed by gel electrophoresis.

Observation

The gel showed the expected bands, confirming successful digestion of both the intermediate recombinant plasmid and the 3′ flank insert.

12th June:

Ligation and Transformation

The digested 3′ flank was ligated into the intermediate recombinant plasmid to obtain the **final recombinant plasmid**.

The ligation product was transformed into competent *E. coli* cells.

Transformation procedure

For the ligation transformation, **100 µl competent cells** were taken and approximately **1 µl DNA (~30 ng)** was added by gentle swirling.

For the pure plasmid transformation, **1 µl plasmid** was added to **100 µl competent cells**.

The cells were:

1. Incubated on ice for **30 min**.
2. Heat-shocked at **42°C for 90 s**.
3. Returned to ice for **5 min**.
4. Mixed with **1 ml sterile LB**.
5. Incubated at **37°C for 45 min** in a shaker incubator.

The transformed cells were subsequently plated on selective medium for colony formation.

Genomic DNA extraction

Genomic DNA extraction of *S. cerevisiae* was also performed.

Conclusion

The ORF was successfully cloned into p426GPD to obtain an intermediate recombinant plasmid. The 3′ flank was then successfully digested and ligated into the intermediate construct, followed by transformation to obtain the final recombinant plasmid.

Week 5: 15th June to 19th June

Agenda

Repetition of the recombinant plasmid construction experiment.

Work Done

The experiment was repeated during the week.

Observation

The experiment was unsuccessful due to contamination of the plate.

Conclusion

The experiment was repeated but was unsuccessful as the entire plate was contaminated.

Week 6: 22nd June to 26th June

Agenda

Repetition of the recombinant plasmid construction experiment.

Work Done

The experiment was repeated during the week.

Observation

The experiment was unsuccessful due to contamination of the plate.

Conclusion

The repeated experiment was unsuccessful as the entire plate was contaminated.

Week 7: 29th June to 3rd July

1st July:

Primary inoculation of the intermediate recombinant plasmid

Two colonies were picked and inoculated separately for plasmid isolation the following day.

- A single colony was inoculated into 5 ml LB broth containing ampicillin. This was repeated for a second colony.
- Both tubes were incubated overnight at 37°C with shaking.

2nd July:

Plasmid isolation of the intermediate recombinant plasmid

Plasmid isolation was performed from both overnight cultures of E. coli containing the intermediate recombinant plasmid.

I. Cell Lysis

1. The bacterial culture was centrifuged at 13,000 rpm for 2 min to pellet the cells.
2. The supernatant was discarded.
3. The pellet was thoroughly resuspended in 250 µl Buffer S1 containing RNase A.
4. 250 µl Buffer S2 was added and the tube was gently inverted 4–5 times without vortexing.
5. The sample was incubated until the lysate became clear, ensuring that the incubation did not exceed 5 min.
6. 350 µl Buffer S3 was added and the tube was immediately inverted 4–6 times without vortexing.
7. The lysate was centrifuged at 13,000 rpm for 10 min.

II. DNA Purification

8. The cleared supernatant was carefully transferred to a mini Q column placed in a collection tube.
9. The column was allowed to stand for 5 min to maximize DNA binding.
10. The column was centrifuged at 13,000 rpm for 30 s and the flow-through was discarded.
11. 500 µl Buffer AW was added, followed by centrifugation for 30 s. The flow-through was discarded.
12. 700 µl Buffer PW was added, followed by centrifugation for 30 s. The flow-through was discarded.

13. The empty column was centrifuged at 13,000 rpm for 1 min to remove residual wash buffer.
14. The spin column was transferred to a fresh 1.5 ml microcentrifuge tube.
15. 30 µl AMQ was added directly to the centre of the membrane without touching the membrane bed.
16. The column was incubated for 5 min at room temperature.
17. The column was centrifuged at 13,000 rpm for 1 min to elute the plasmid DNA.

Restriction digestion

Restriction digestion was performed on:

Vector: Intermediate recombinant plasmid

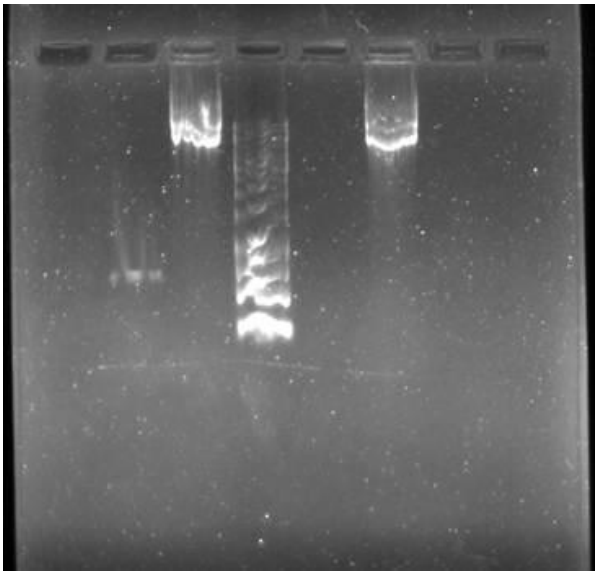
Insert: 3′ flank

The restriction enzymes SacI and BamHI were used. The digestion was performed sequentially, with SacI digestion followed by heat inactivation and subsequent BamHI digestion.

Component	**Volume**
DNA template	
10X NEB buffer	
SacI	
BamHI	
AMQ	
Total volume	

Gel extraction and purification

The digested products were run on a gel.



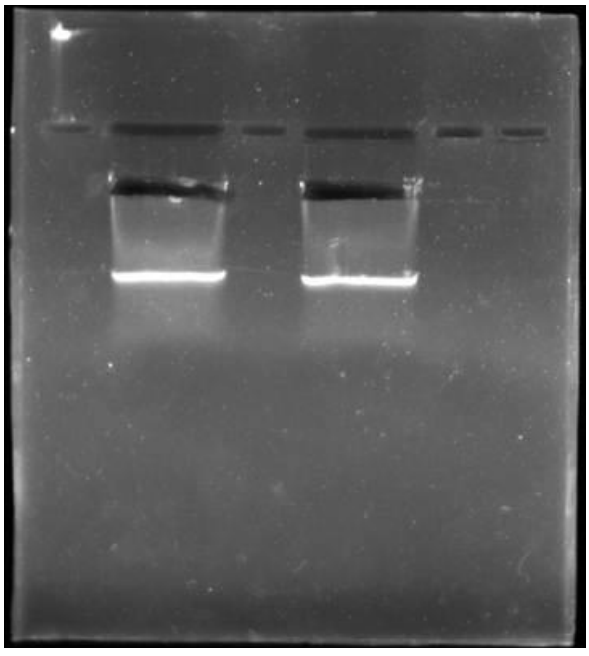
Observation

[yet to add]

The bands corresponding to the digested intermediate recombinant plasmid and the 3′ flank insert were excised from the gel for purification and thus extraction.

Purification

18. 3 volumes of GB Buffer were added to the excised gel slice.
19. The solution was mixed thoroughly until it was homogeneous.
20. The mixture was transferred to a spin column placed inside its collection tube.
21. The column was centrifuged at 13,000 rpm for 1 min.
22. The flow-through was discarded and the spin column was placed back into the collection tube.
23. 700 µl of NW Buffer was added to the spin column.
24. The column was centrifuged at 13,000 rpm for 30 s.
25. The flow-through was discarded and the spin column was returned to the collection tube.
26. An additional centrifugation was performed at 13,000 rpm for 1 min to remove residual wash buffer.
27. The spin column was transferred to a fresh 1.5 ml microcentrifuge tube.
28. 30 µl of AMQ was added directly to the centre of the membrane, taking care not to touch the membrane with the pipette tip.
29. The column was incubated at room temperature for 10 min.
30. The column was centrifuged at 13,000 rpm for 1 min.
31. The purified DNA was collected in the microcentrifuge tube.



Observation

The required bands were excised from the gel for purification.

Media and material preparation

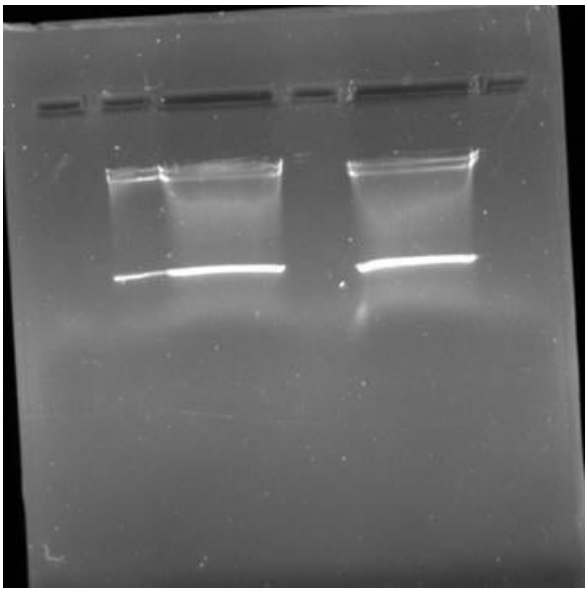
The following were autoclaved:

- 20 test tubes
- 8 spreaders
- 4 LA plates
- 4 LA + ampicillin plates

3rd July:

Gel electrophoresis and gel extraction

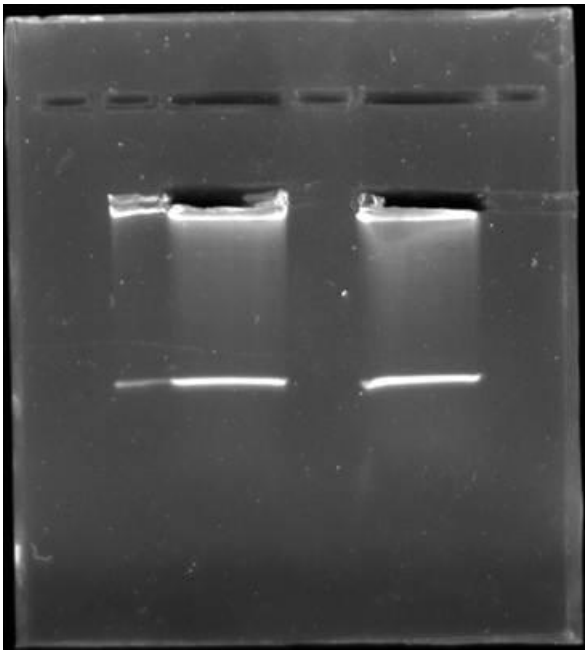
A gel electrophoresis run was performed on the digested vector.



Observation

Two bands were resolved in each lane.

The upper band was excised from both lanes for purification since those contained our plasmid.

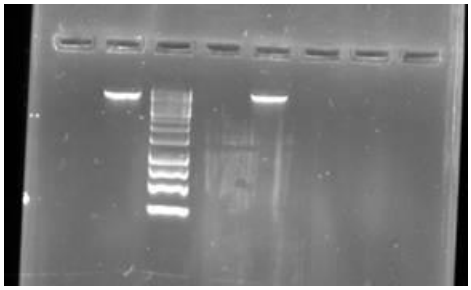


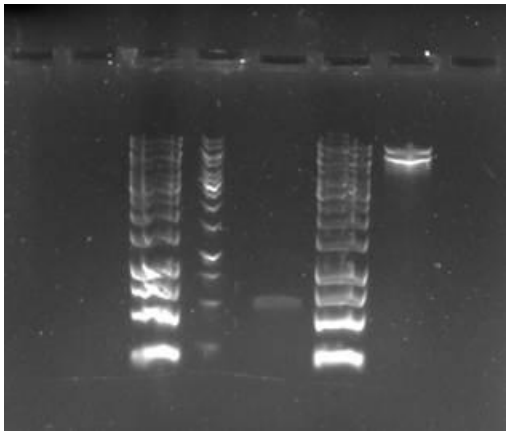
Purification

1. 3 volumes of GB Buffer were added to the excised gel slices.
2. The solution was mixed thoroughly until it was homogeneous.
3. The mixture was transferred to a spin column placed inside its collection tube.
4. The column was centrifuged at 13,000 rpm for 1 min.
5. The flow-through was discarded and the spin column was placed back into the collection tube.
6. 700 µl of NW Buffer was added to the spin column.
7. The column was centrifuged at 13,000 rpm for 30 s.
8. The flow-through was discarded and the spin column was returned to the collection tube.
9. An additional centrifugation was performed at 13,000 rpm for 1 min to remove residual wash buffer.
10. The spin column was transferred to a fresh 1.5 ml microcentrifuge tube.
11. 30 µl of AMQ was added directly to the centre of the membrane, taking care not to touch the membrane with the pipette tip.
12. The column was incubated at room temperature for 10 min.
13. The column was centrifuged at 13,000 rpm for 1 min.
14. The purified DNA was collected in the microcentrifuge tube.

Gel electrophoresis after purification

A gel electrophoresis run was performed to check the purified vector and the purified 3′ flank insert.





Observation

A single band was obtained for the purified vector, confirming successful gel extraction and purification.

Ligation

The purified 3′ flank insert was ligated into the intermediate recombinant plasmid to obtain the final recombinant plasmid.

The ligation reaction consisted of:

Component	**Volume**
Vector (intermediate recombinant plasmid)	
Insert (3′ flank)	
10X ligase buffer	
DNA ligase	
AMQ	
Total volume	

The ligase buffer was completely thawed and vortexed to ensure that the salts were properly mixed. After preparing the ligation mixture, the tube was gently tapped to mix and briefly centrifuged. The reaction was incubated for [incubation time to be added].

Transformation into competent E. coli

1. The ligation was set up in a microcentrifuge tube and 100 µl of competent cells was added after thawing. The mixture was swirled to mix.
2. The tube was left on ice for 30 min.
3. A heat shock was given on a heat block for exactly 90 s.
4. The tube was kept on ice for 5 min.
5. 1 ml of sterile LB was added and the tube was taped to a shaker incubator for 45 min.
6. The tube was centrifuged at 13,000 rpm for 2 min.
7. The supernatant was discarded.
8. 100 µl of AMQ was added and the pellet was vortexed until resuspended.
9. The suspension was plated on an LA + ampicillin plate.
10. The plate was placed in the incubator overnight at 37°C.

Week 8: 6th July to 10th July

Agenda

- Plasmid isolation and screening of the final recombinant plasmid
- Restriction digestion screening to confirm construction of the final recombinant plasmid
- Confirmation that the 3′ flank is present in the final recombinant plasmid and absent from the intermediate recombinant plasmid
- Linearisation of the confirmed final recombinant plasmid by restriction digestion
- Transformation of S. cerevisiae with the linearised plasmid

Instruments required

- PCR machine
- Water bath
- Centrifuge
- Shaker incubator
- Gel electrophoresis apparatus

- UV transilluminator
- Micropipettes
- PCR tubes and microcentrifuge tubes

Reagents

- AMQ
- 10X PCR buffer
- 1 mM dNTPs
- GAP promoter forward primer and reverse primer
- PCR enzyme
- Restriction enzymes SacI, BamHI, EcoRI and SmaI
- 10X NEB buffer
- Agarose
- AZ dye
- TBE buffer
- GeneRuler 1 kb ladder
- LB medium and LA + ampicillin plates
- YPD medium
- SD-URA plates

6th July:

Primary inoculation

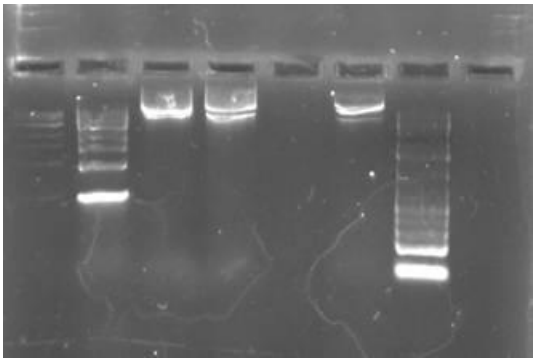
Colonies from the transformation plate were inoculated into 5 ml LB broth containing ampicillin and incubated overnight at 37°C with shaking.

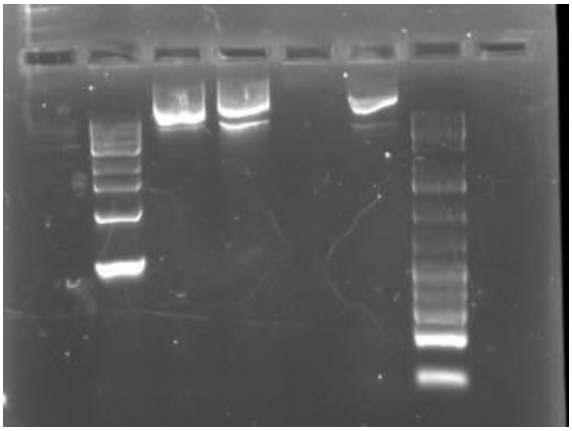
Plasmid isolation of the final recombinant plasmid

Plasmid isolation was performed from the overnight cultures, following the same procedure done on 2nd July.

Gel electrophoresis

Two gel electrophoresis runs were performed on the isolated circular plasmids.





Observation

[yet to add]

7th July:

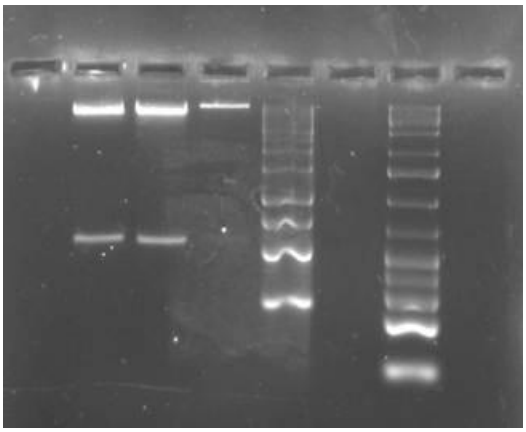
Restriction digestion screening

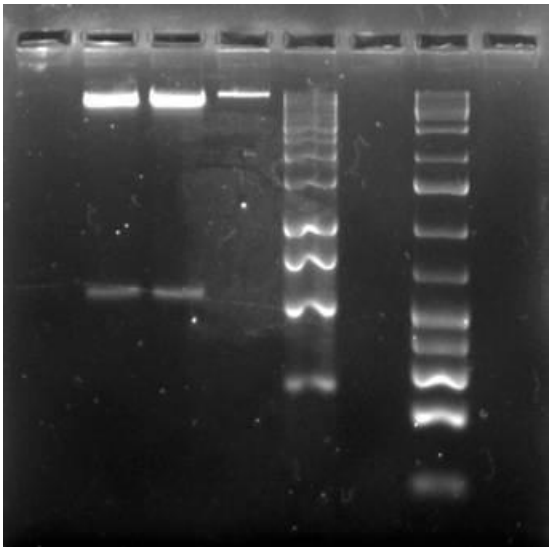
Restriction digestion was performed on the isolated plasmids to check for the presence of the final recombinant construct.

Component	**Volume**
DNA template	
10X NEB buffer	
SacI	
BamHI	
AMQ	
Total volume	

Gel electrophoresis

The digested plasmids were resolved by gel electrophoresis. A long run was also performed to improve band separation.





Observation

The screening was successful. Two distinct bands corresponding to the vector and the insert were observed, confirming construction of the final recombinant plasmid.

8th July:

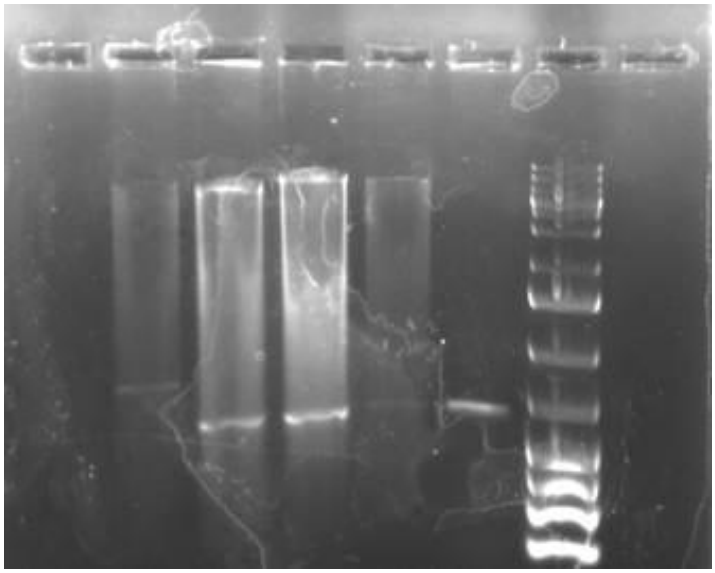
Restriction digestion of the recombinant plasmids

Restriction digestion was performed on recombinant plasmids 1, 2 and 3 and on the intermediate recombinant plasmid, to check whether the 3′ flank was present in the final recombinant plasmid and absent from the intermediate recombinant plasmid. The restriction enzymes SacI and BamHI were used.

Component	Volume
AMQ	6 µl
10X buffer	1 µl
SacI	0.5
BamHI	0.5
DNA	2 µl
Total	10

Gel electrophoresis

The digested products were resolved by gel electrophoresis alongside the ORF and the GeneRuler 1 kb ladder.



Observation

[yet to add]

PCR amplification

A gradient PCR was set up using the GAP promoter forward primer with the reverse primer, to check for the presence of the 3′ flank in the final recombinant plasmid. The expected amplicon size was 2943 bp.

The reaction mix was prepared as a pool for three reactions:

Component	Per reaction	Pool (×3)
AMQ	10 µl	30 µl
10X buffer	2 µl	6 µl
1 mM dNTPs	2 µl	6 µl
5 µM forward primer (GAP)	2 µl	6 µl
5 µM reverse primer	2 µl	6 µl
Template (plasmid, 1:40)	2 µl	Added separately to each reaction
Enzyme	0.2 µl	0.6 µl

Templates used: ORF, final recombinant plasmid 1 and final recombinant plasmid 2.

Cycling conditions:

- Initial denaturation: 98°C, 5 min
- Denaturation: 98°C, 30 s
- Annealing: Tm, 30 s
- Extension: 72°C, 3 min
- Steps 2 to 4 repeated for 20 cycles
- Final extension: 72°C, 5 min
- Hold: 4°C, 2 min

The recorded primer annealing temperatures were 43°C for the GAP promoter primer and 57°C for the reverse primer.

Observation

The PCR was unsuccessful. The difference between the annealing temperatures of the two primers was too large for a single annealing temperature to work for both.

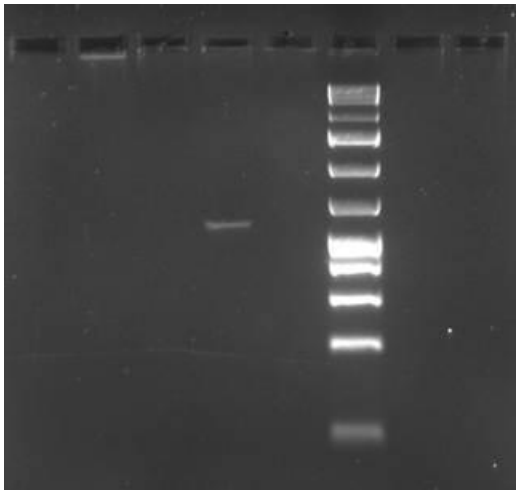
9th July:

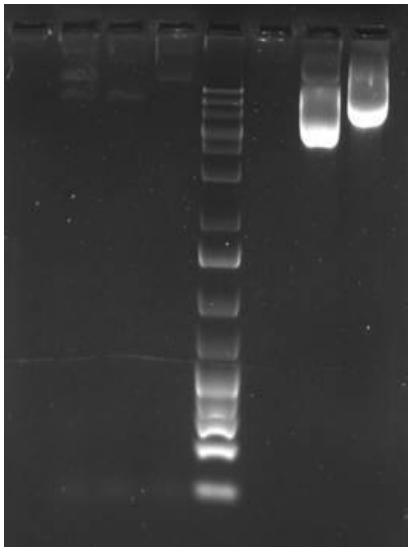
PCR amplification (repeated)

The PCR was repeated at an annealing temperature of 65°C to check for the presence of the 3′ flank in the final recombinant plasmid and its absence from the intermediate recombinant plasmid.

Gel electrophoresis

The PCR products were resolved by gel electrophoresis alongside the GeneRuler 1 kb ladder.





Observation

The 3′ flank PCR was successful, confirming that the 3′ flank was present in the final recombinant plasmid.

Primary inoculation of S. cerevisiae

A primary inoculation of S. cerevisiae was performed in 5 ml YPD and incubated overnight at 30°C with shaking, in preparation for yeast transformation.

10th July:

Secondary inoculation of S. cerevisiae

A secondary inoculation was set up from the overnight primary culture and grown until the culture reached the required optical density for transformation i.e. near 0.8 to 1.

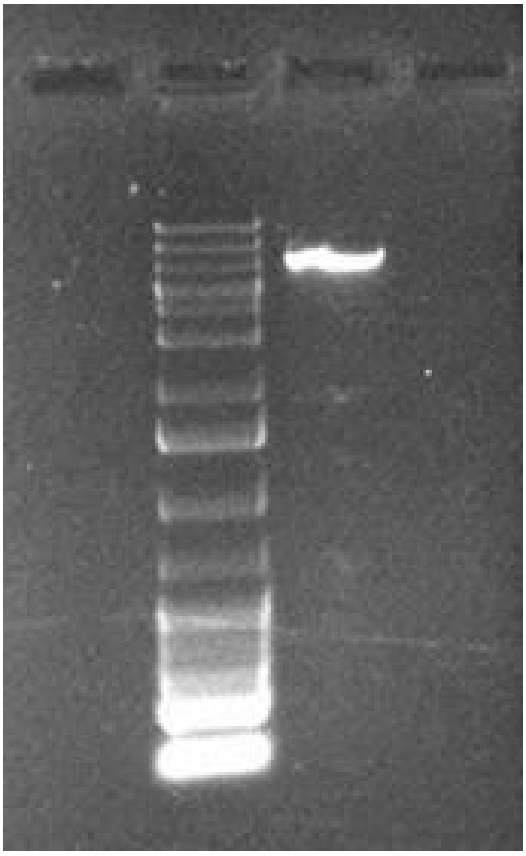
Linearisation of the final recombinant plasmid

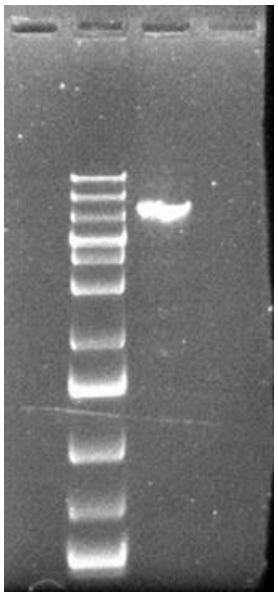
The confirmed final recombinant plasmid was linearised by restriction digestion using EcoRI and SmaI, in preparation for integration into the yeast genome by homologous recombination.

Component	**Volume**
DNA template	
10X NEB buffer	
EcoRI	
SmaI	
AMQ	
Total volume	

Gel electrophoresis

The digested plasmid was resolved by gel electrophoresis to confirm linearisation. A long run was also performed.





Observation

A single band was observed for the digested plasmid, confirming that the final recombinant plasmid had been linearised.

Yeast transformation

S. cerevisiae was transformed with the linearised final recombinant plasmid. The transformation was carried out with leucine.

1. The inoculated cells were pelleted by centrifugation at 13,000 rpm for 15 s.
2. 500 µl AMQ was added to each and the cells were resuspended by vortexing.
3. Both were pooled into one microcentrifuge tube.
4. The tube was centrifuged for 15 s at 13,000 rpm.
5. 1 ml LiAc (100 mM) was added and the pellet was resuspended.
6. The tube was taped to the shaker incubator at 200 rpm, 30 °C, for 1.5 h. During this time ssDNA was boiled for 10 min and then placed on ice.
7. 200 µl of the cell suspension was taken into a new microcentrifuge tube.
8. The tube was centrifuged for 15 s at 13,000 rpm.
9. The supernatant was discarded, leaving a little volume for resuspension.
10. The following were added in order: 240 µl PEG, 36 µl 1 M LiAc, 20 µl ssDNA (2 mg/ml), and the linearised plasmid. The pipette tip was not put into the mix; the reagents were poured in from above.
11. The mixture was mixed by vortexing.
12. The tube was taped to the shaker incubator at 200 rpm, 30 °C, for 1 h.
13. 45 µl DMSO was added and mixed by inverting the tube with a light vortex.
14. The tube was kept in a water bath at 42 °C for 30 min for heat shock.
15. The tube was centrifuged and the supernatant was discarded.
16. The cells were resuspended in 1 ml YPD for recovery and incubated for 3 h.
17. The tube was centrifuged at 13,000 rpm for 15 s.
18. The cells were washed with 1 ml sterile AMQ.

19. The cells were resuspended in 100 µl sterile AMQ and plated on selection media.
20. The plates were incubated at 30 °C for 2 to 3 days.

Conclusion

The final recombinant plasmid was isolated and screened by restriction digestion, which gave two distinct bands corresponding to the vector and the insert and confirmed that the construct had been assembled correctly. The first attempt to confirm the 3′ flank by PCR failed because the annealing temperatures of the GAP promoter primer and the reverse primer differed by 14°C, which is too large a gap for a single annealing temperature. Repeating the reaction at a revised annealing temperature gave the expected product, confirming that the 3′ flank was present in the final recombinant plasmid. The confirmed plasmid was then linearised with EcoRI and SmaI and used to transform S. cerevisiae.

Week 9: 13th July to 19th July

Agenda

● Transformation of *S. cerevisiae* BY4742 with the linearised final recombinant plasmid
● Selection of transformants using the appropriate auxotrophic marker
● Patching of colonies from the transformation plate
● Primary inoculation of selected patches for further screening

Yeast strain: *Saccharomyces cerevisiae* BY4742

Auxotrophy markers

● HIS3
● LYS2
● URA2

Instruments required

● Centrifuge
● Shaker incubator
● Water bath
● Vortex mixer
● Micropipettes
● Microcentrifuge tubes

● Test tubes
● Incubator

Reagents

● YPD medium
● AMQ
● 100 mM LiAc
● PEG
● ssDNA
● Linearised final recombinant plasmid
● DMSO
● SD-URA medium/plates
● Lysine-containing SD-URA medium/plates

13th July:

Review of transformation setup

The transformation carried out on 10th July used leucine for selection. It was later realised that the selection condition was incorrect, so the existing setup was discontinued and the selection condition was changed for the next transformation.

14th July:

SD-URA plating and primary inoculation

Plating was carried out on SD-URA medium with the appropriate lysine-related selection condition. A primary inoculation of *S. cerevisiae* BY4742 was also prepared for the upcoming transformation.

15th July:

Transformation with the final recombinant plasmid

S. cerevisiae BY4742 was transformed with the linearised final recombinant plasmid and plated on SD-URA medium using the selected conditions. A fresh primary inoculation was also prepared in case the transformation needed to be repeated.

Observation

Around 15 colonies started appearing overnight on the transformation plate which is faster than expected. This raised some doubts about whether the transformation had worked as expected.

16th July:

Repeat transformation

The culture that had been primary-inoculated earlier was used for another transformation. A fresh primary inoculation was also prepared so that another culture would be available if required.

17th July:

Transformation and patching

The culture inoculated on 16th July was used for another transformation. The colonies from the 15th July transformation plate were also patched into 30 individual patches for further screening.

18th July:

The laboratory was closed, so no experimental work was carried out.

19th July:

The laboratory was closed, so no experimental work was carried out.

Conclusion

The transformation experiments were repeated after an issue with the initial selection condition was identified. A total of 30 patches were prepared from the 15th July transformation plate and were used for further screening in the following week.

Week 10: 20th July to 24th July

Agenda

● Primary inoculation of the 30 patches
● gDNA extraction from selected patches
● Gel electrophoresis of extracted gDNA
● Repeat extraction and screening of selected samples

Instruments required

● Centrifuge
● Gel electrophoresis apparatus
● Micropipettes
● Microcentrifuge tubes
● Vortex mixer
● Water bath
● Shaker incubator
● Incubator

Reagents

gDNA extraction

● AMQ
● Triton-SDS
● Phenol-chloroform
● TE buffer, pH 8.0
● Glass beads
● Absolute ethanol
● 70% ethanol
● 4 M ammonium acetate

Gel electrophoresis

● Agarose
● TBE buffer
● AZ/DNA stain
● DNA ladder

gDNA extraction from *S. cerevisiae*

Cell preparation

1. Centrifuge the culture at 13,000 rpm for 15 seconds.
2. Remove the supernatant and add 1 mL sterile AMQ.
3. Resuspend the cells by gentle vortexing.

4. Centrifuge at 13,000 rpm for 15 seconds.
5. Remove the supernatant and resuspend the cells in 1 mL sterile water.
6. Centrifuge again at 13,000 rpm for 15 seconds at room temperature and remove the supernatant.

DNA extraction

7. Add 0.2 mL Triton-SDS and resuspend the pellet.
8. Add 0.2 mL phenol-chloroform and 0.2 g glass beads.
9. Vortex for 30 seconds at room temperature.
10. Add 0.2 mL TE buffer (pH 8.0) and vortex for another 30 seconds.
11. Centrifuge at 13,000 rpm for 10 minutes to separate the phases.
12. Transfer the aqueous layer to a fresh microcentrifuge tube without disturbing the interface.

DNA isolation

13. Add 0.7 mL chilled absolute ethanol and 10 µL 4 M ammonium acetate.
14. Invert the tube 5–10 times to mix.
15. Incubate at −20°C for 12 minutes.
16. Centrifuge at 13,000 rpm for 10 minutes.
17. Discard the supernatant.
18. Wash the DNA pellet with 0.5 mL 70% ethanol.
19. Centrifuge at 13,000 rpm for 10 minutes and discard the ethanol.
20. Dry the DNA pellet according to the documented protocol.
21. Dissolve the pellet in 30 µL AMQ.

gDNA gel electrophoresis

The extracted gDNA samples were run on an agarose gel using TBE buffer, AZ/DNA stain and a DNA ladder.

20th July:

Primary inoculation of patches

All 30 patches from the 15th July transformation plate were individually inoculated for further screening.

21st July:

gDNA extraction

gDNA was extracted from patches 1, 2, 3 and 4. The samples were labelled as vials 1–4 and kept for gel electrophoresis.

22nd July:

Gel electrophoresis

The gDNA samples from vials 1–4 were run on an agarose gel. The result was negative, so additional patches were selected for screening.

Primary inoculation of additional patches

Patches 27, 5, 6, 7, 8, 9 and 26 were primary-inoculated for further analysis.

23rd July:

gDNA extraction and repeat screening

gDNA was extracted from the newly inoculated patches, and the samples were run on a gel. The result was unsuccessful, so gDNA extraction was repeated for samples 5, 8 and 26.

24th July:

Gel electrophoresis

The repeated gDNA samples from patches 5, 8 and 26 were run on an agarose gel. The gel run was successful.

Laboratory cleaning

The laboratory was cleaned after the experimental work was completed.

Conclusion

All 30 patches were inoculated during Week 10, and selected samples were screened through gDNA extraction and gel electrophoresis. The first gel run using samples 1–4 was unsuccessful. Further samples were then tested, and gDNA extraction was repeated for patches 5, 8 and 26. The final gel run on 24th July was successful, completing the screening work for the week.

Week 11 : 11th August to 13th August

11th August 2026

Primer Design for Confirmation of Yeast Transformation

Primers were designed to test whether the transformation and subsequent **homologous recombination** in engineered *S. cerevisiae* had been successful.

Two primer pairs were designed:

Target	Primer	Location	Forward Primer Tm	Reverse Primer Tm	Tm used	Expected amplicon
3′ flank	A, B	A – plasmid sequence; B – *S. cerevisiae* genomic sequence	61°C	62°C	61°C	2.5 kb
ORF	C, D	C – plasmid sequence; D – *S. cerevisiae* genomic sequence	61°C	59°C	60°C	4 kb

The primer arrangement was designed such that amplification would provide evidence for the presence of the corresponding recombinant region in the yeast genome.

Before testing the primers on the engineered yeast samples, it was necessary to confirm that the primers themselves were functional. Therefore, the primers were first tested using known DNA templates.

Primer Functionality Test

For the initial test:

- **Primers A and D** were tested using the plasmid as the template.
- **Primers B and C** were tested using *S. cerevisiae* gDNA as the template.

The PCR setup and cycling conditions were prepared as recorded in the experimental setup shown below.

12th August 2026

Gel Electrophoresis

A gel run was performed to check the PCR products obtained from the primer functionality tests.

The PCR using **primer A** gave the expected confirmation, indicating that primer A was functional.

The other PCR reactions did not give the required amplification. Therefore, the remaining primer sets were planned for **optimization and repetition** before being used for testing the engineered yeast samples.

Observation

- Primer A: PCR successful.
- Other primer tests: PCR unsuccessful.
- Further PCR optimization required.

13th August 2026

PCR Optimization and Repetition

The PCR reactions for the remaining primers were repeated after optimization.

The PCR for the remaining primer sets was successful, confirming that the primers could be used for the subsequent testing of the engineered yeast samples.

Observation

The remaining primer PCRs worked successfully after optimization and repetition.

Conclusion

The primer sets designed to confirm successful homologous recombination were tested for functionality before being used on engineered yeast samples. Primer A initially worked, while the remaining reactions required optimization; after repetition, the remaining primer PCRs were successful.

17th August 2026

Agenda

PCR-based confirmation of the engineered *S. cerevisiae* sample using genomic DNA and control templates.

Procedure

PCR was performed to check the genomic DNA of engineered *S. cerevisiae* sample 5 (dated 8/26, as recorded). The final recombinant plasmid and genomic DNA were also used as controls for comparison.

Observations

- An amplicon was observed in sample 5.
- The DNA ladder did not resolve completely, making it difficult to determine the size of the amplicon.
- Therefore, although amplification was observed in sample 5, the amplicon size could not be confirmed.

Conclusion

PCR amplification was observed in engineered *S. cerevisiae* sample 5. However, incomplete resolution of the DNA ladder prevented confirmation of the amplicon size.

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Drylab results:
Figuring out mutation site

V1254W MUTAION: plasmid sequencing + gel runs all
Fba for growth rate degradation prediction
GCMS

1. CFD results: simulation+conclusion
2. LCA result: comparative environmental impact. Our is best

DBTL
Dbtl 3 : 3 flank
Dbtl 4 : transformation into yeast
Dbtl 5 : gcms.